DEVELOPER ROLE

Answer Key — 100 Questions

Keep this file separate from the question paper during practice or assessment.

1. B — Encapsulation

2. C — final

3. C — main()

4. A — 0

5. C — HashSet

6. B — ArrayList

7. A — Same method name with different parameter lists

8. B — Providing a subclass implementation of an inherited method

9. C — FileNotFoundException

10. B — this

11. B — HashMap

12. B — Java Virtual Machine

13. A — A class can implement multiple interfaces

14. B — One class-level member shared by instances

15. C — String

16. B — Initialize a newly created object

17. D — public

18. B — finally

19. B — Logical equality of objects

20. C — throw

21. A — O(1)

22. B — Stack

23. C — Queue

24. C — O(1)

25. B — Breadth-first search

26. C — Inorder

27. B — O(log n)

28. A — Merge sort

29. B — Stack

30. D — $O(n^2)$

31. A — Dynamic programming

32. B — O(V + E)

33. C — Priority queue

34. A — O(1)

35. B — Floyd's slow and fast pointers

36. B — Keeps search, insertion, and deletion around O(log n)

37. B — SELECT

38. C — WHERE

39. B — HAVING

40. C — Primary key

41. B — Referential integrity

42. D — INNER JOIN

43. B — All rows in the result set

44. B — UPDATE

45. C — Atomicity

46. B — Redundancy and update anomalies

47. A — B-tree/B+ tree

48. B — Aggregate rows sharing the same grouping values

49. B — TRUNCATE

50. B — A logical unit of work that is committed or rolled back

51. C — GET

52. B — 201

53. B — 401

54. B — An architectural style for networked applications

55. A — application/json

56. B — Translate domain names to IP addresses

57. B — TCP

58. A — TLS-based transport security

59. D — 500

60. B — Business/service layer

61. B — Cross-origin browser requests

62. B — PATCH

63. B — Return large result sets in manageable chunks

64. A — @RestController

65. B — Providing an object with its required dependencies externally

66. C — @GetMapping

67. B — Transfer data between application boundaries

68. A — Controller

69. B — Abstract data access operations

70. A — Database transaction boundaries

71. B — application.properties

72. B — Version and apply database schema changes

73. C — A slow salted password hash such as BCrypt or Argon2

74. B — Help verify token integrity and authenticity

75. C — Secret manager or protected environment configuration

76. B — An attack that manipulates SQL through untrusted input

77. B — Using parameterized queries/prepared statements

78. B — Grant only permissions required for the task

79. B — Track changes to files and coordinate version control

80. B — git switch -c branch-name

81. B — Fetches remote changes and integrates them into the current branch

82. B — Competing changes Git cannot automatically reconcile

83. A — Request review and integration of proposed changes

84. B — Frequently integrating changes with automated build and test checks

85. A — Unit testing

86. B — Simulate a dependency and control its behavior

87. B — Checking that existing behavior still works after changes

88. B — Improve correctness, maintainability, and shared understanding

89. A — Scheduler

90. B — A state where processes wait indefinitely for resources held by one another

91. A — Heap

92. B — A unit of execution within a process

93. A — Repeated access latency by keeping reusable data closer to the consumer

94. B — Adding more machines or instances to share workload

95. A — Distribute incoming traffic across backend instances

96. B — Repeating the same request should have the same intended effect

97. A — Logs, metrics, and traces

98. B — Keep deployment settings separate from application code

99. B — Establish reliable reproduction steps and expected versus actual behavior

100. A — Single Responsibility Principle